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Supplementary information

Author Correction: A high-performance neuroprosthesis for speech decoding and avatar control

In the format provided by the authors and unedited

Main article

Text

One sentence in the text of the main article has been updated as part of this correction. With the corrected anatomical assignments for the electrodes, we observed that some temporal-lobe electrodes both 1) were activated by passive listening and 2) contributed to the decoding models for silently attempted speech.

New text is on the top; old text is on the bottom

We found that some temporal-lobe electrodes were not only active during passive listening but also contributed to silently attempted speech decoding ($r >= 0.55$, $P < 0.0001$, Pearson correlation permutation test, Supplementary Fig. 16), suggesting that they may record cortical activity from the subcentral gyrus[36] or sites with production activity within the temporal lobe[37].

Notably, temporal-lobe electrodes contributing to decoding were not strongly activated during auditory perception ($r < 0.1$, $P > 0.01$, Pearson correlation permutation test; Supplementary Fig. 16), suggesting that they may record cortical activity from the subcentral gyrus[36] or production-specific sites within the temporal lobe[37].

Figure 1d

The middle portion of panel d of Figure 1 has been revised to now depict electrode-activation maps using the corrected anatomical arrangement of the electrodes. Articulatory activations for different articulators (jaw, lips, and tongue, shown in brown, blue, and green, respectively) appear more spatially clustered using the corrected anatomical arrangement.

New panel portion is on the top; old panel portion is on the bottom

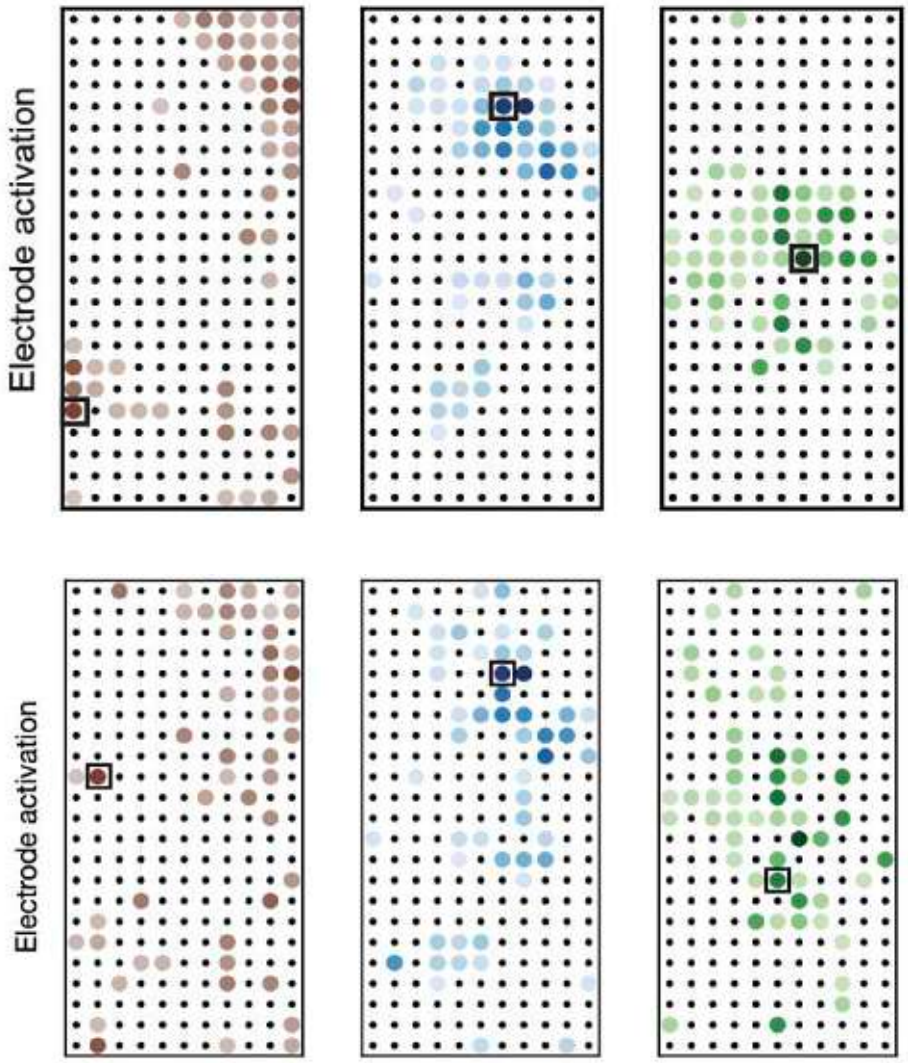
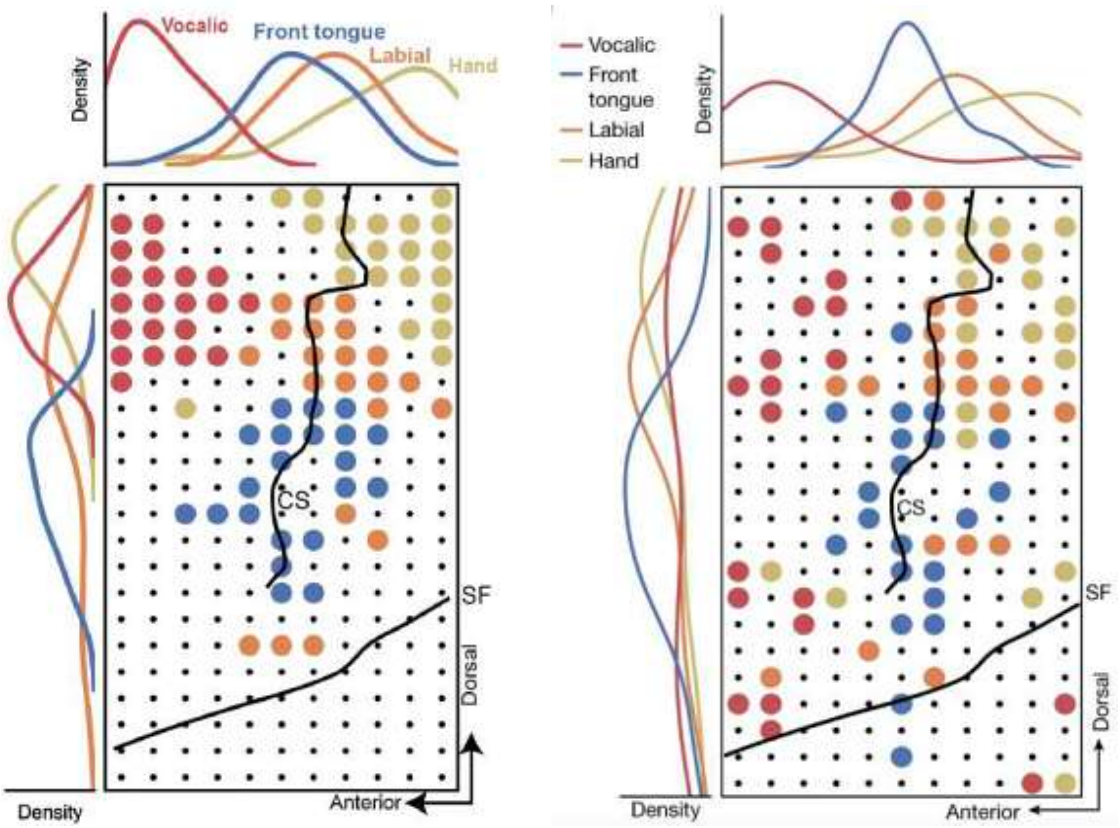


Figure 5f

Panel f of Figure 5 has been revised using the corrected anatomical arrangement of the electrodes. The corrected version depicts clearer spatial clusters of articulatory encoding, which is more consistent with prior literature on somatotopy in the sensorimotor cortex.

New figure is on the left; old figure is on the right

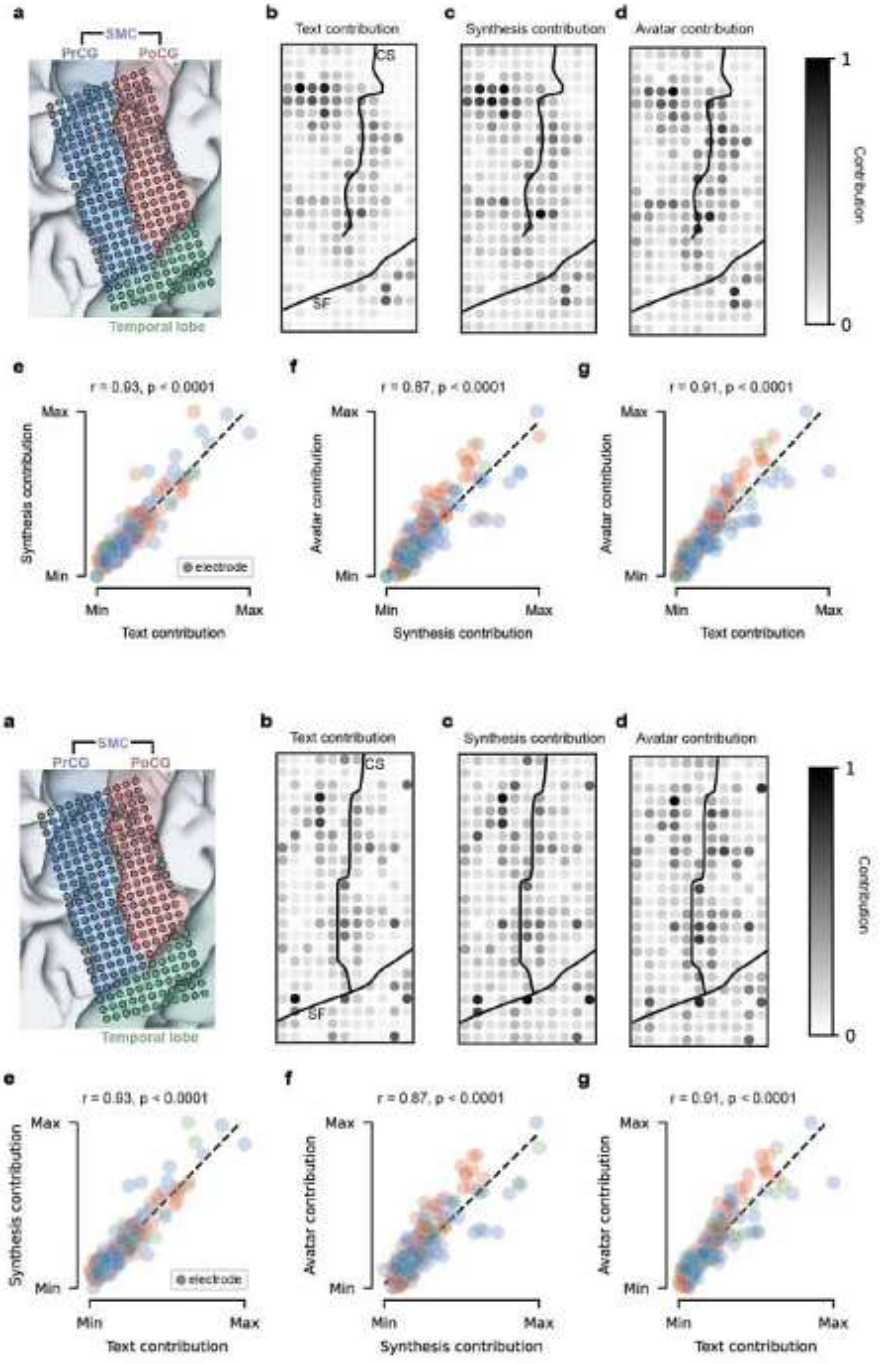


Extended data

Extended data figure 7

(**b**–**d**) Electrode-contribution maps are now depicted in the corrected anatomical arrangement. (**e**–**g**) Electrode anatomical regions (colors) have been corrected; correlations and statistics were unaffected.

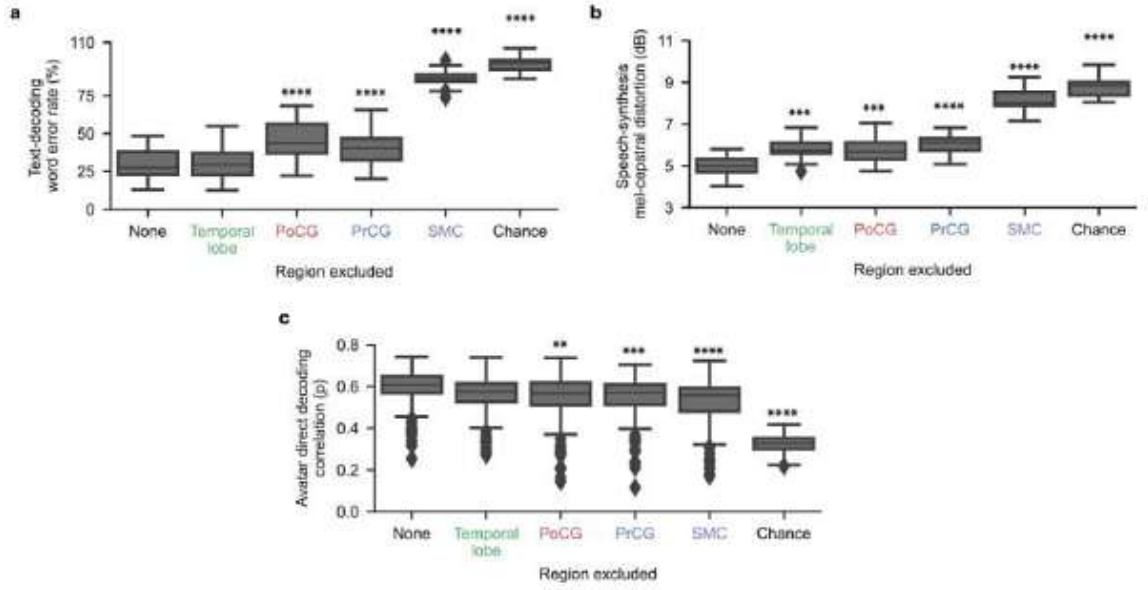
New figure is on the top; old figure is on the bottom

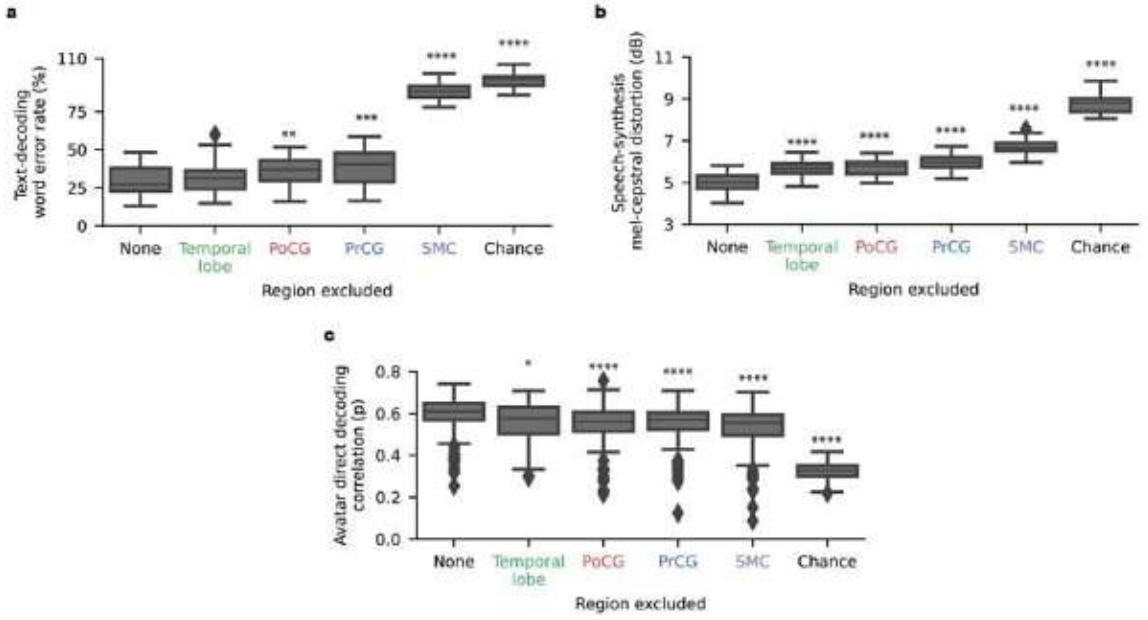


Extended data figure 8

The effects of excluding electrodes from certain anatomical regions on decoding performance have been updated after re-running analyses with the corrected anatomical electrode assignments. Statistically significant comparisons that were present before the correction are still present with the corrected analyses, except that exclusion of the temporal lobe no longer shows a statistically significant effect on avatar direct decoding. Also, the significance levels of some comparisons are slightly different with the corrected analyses.

New figure is on the top; old figure is on the bottom





Supplementary information

Figure S8

Electrode encoding magnitudes are now depicted in the corrected anatomical arrangement.

New figure is on the left; old figure is on the right

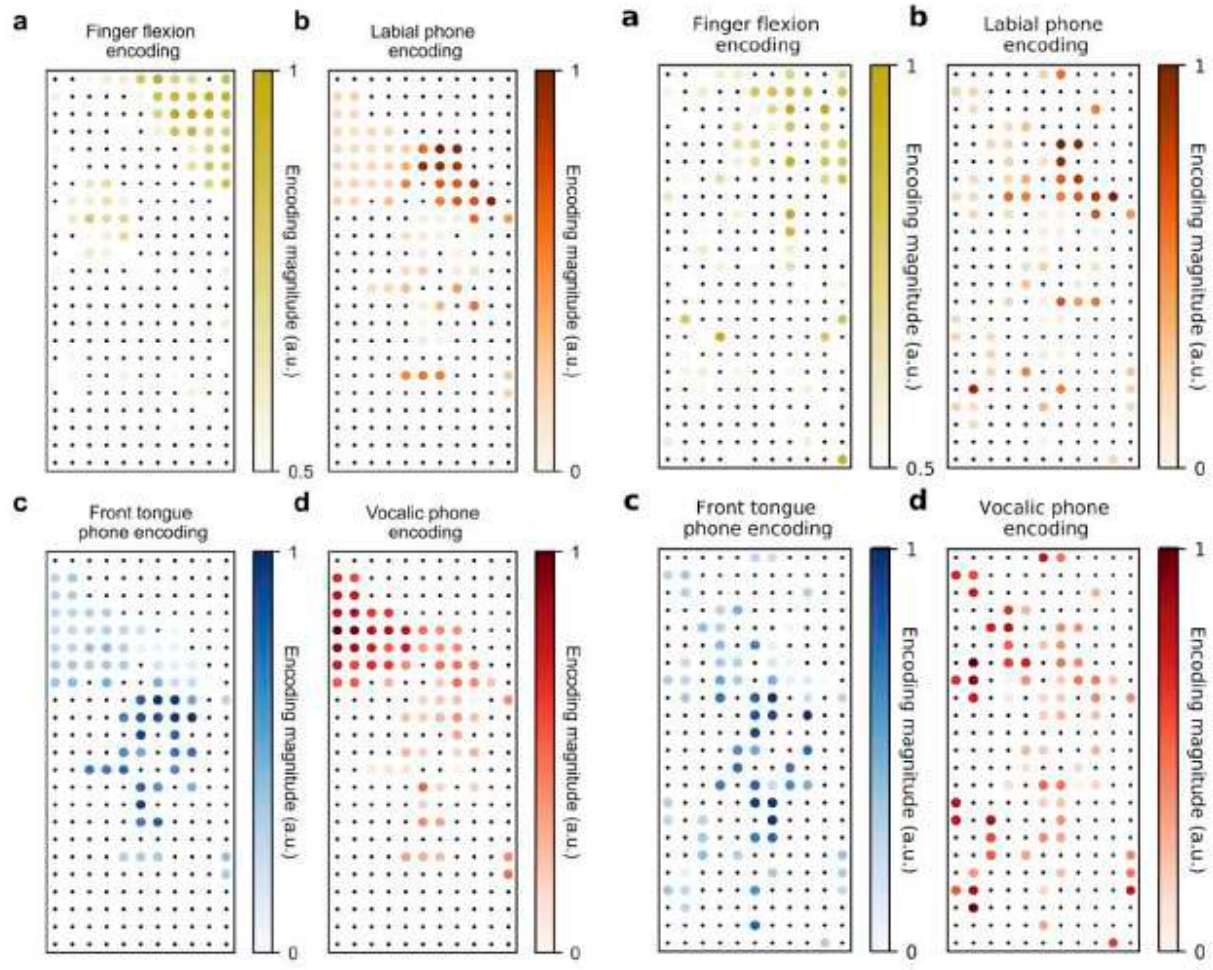
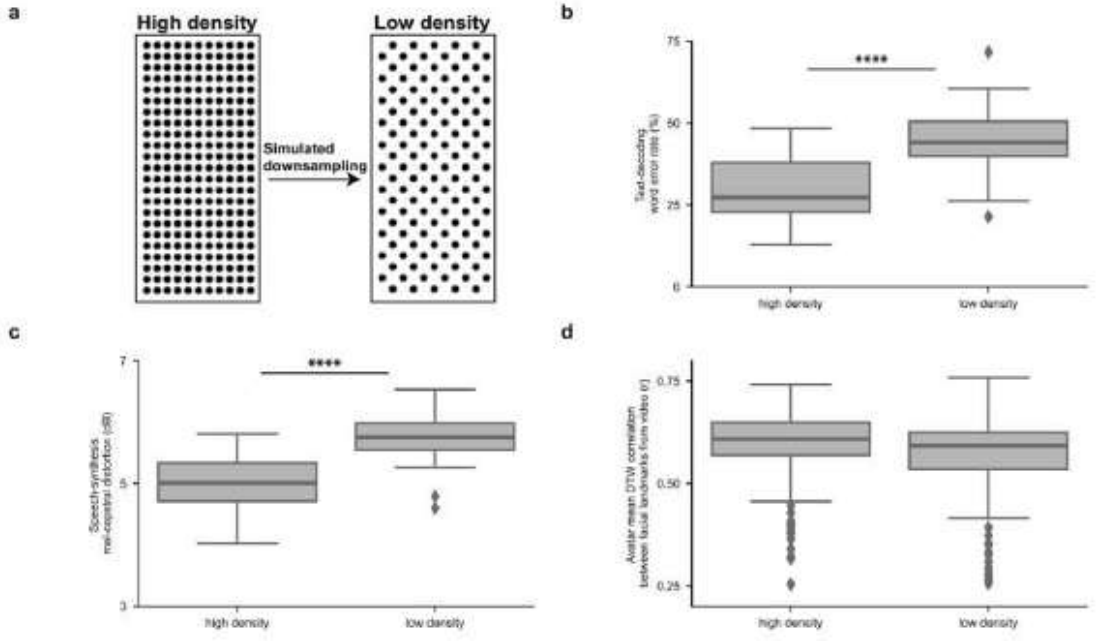


Figure S10

The effects of simulating a lower-density cortical sampling on decoder performance have been updated after re-running analyses with the corrected anatomical electrode arrangements. Results are generally similar to the outcomes before the correction, with the most notable difference being that the effect on avatar facial-landmark correlations is no longer statistically significant (although the median correlation is still lower when using a lower-density sampling).

New figure is on the top; old figure is on the bottom



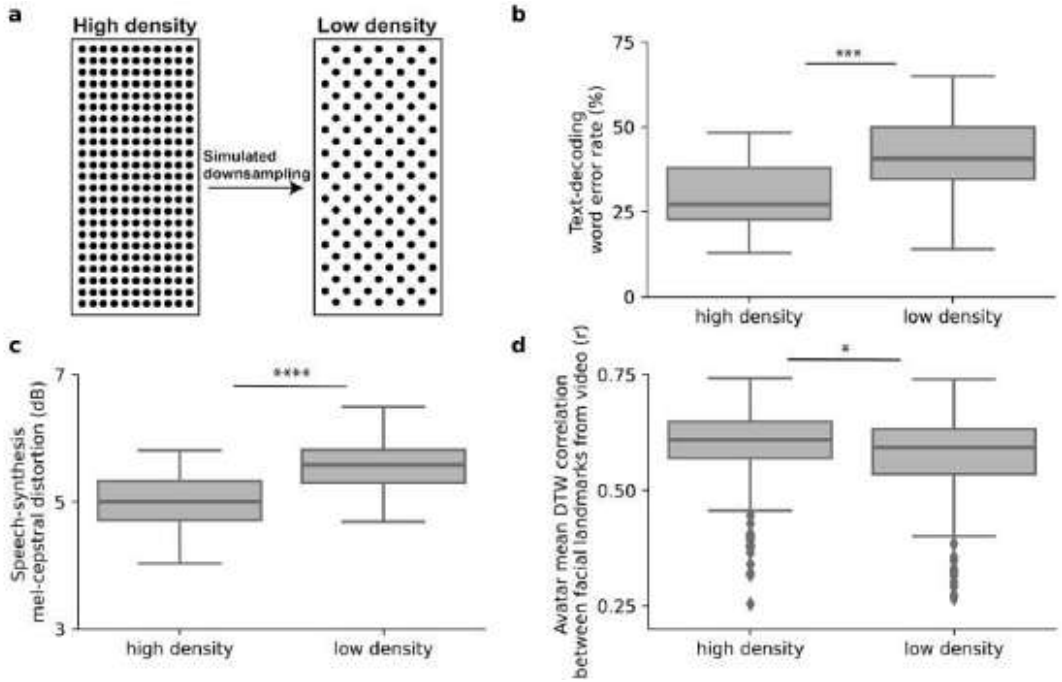


Figure S11

Phone confusion matrices computed using the outputs from the text decoder while including all electrodes and while including a simulated lower-density sampling of electrodes have been updated after re-running analyses with the corrected anatomical electrode arrangements.

New figure is on the left; old figure is on the right

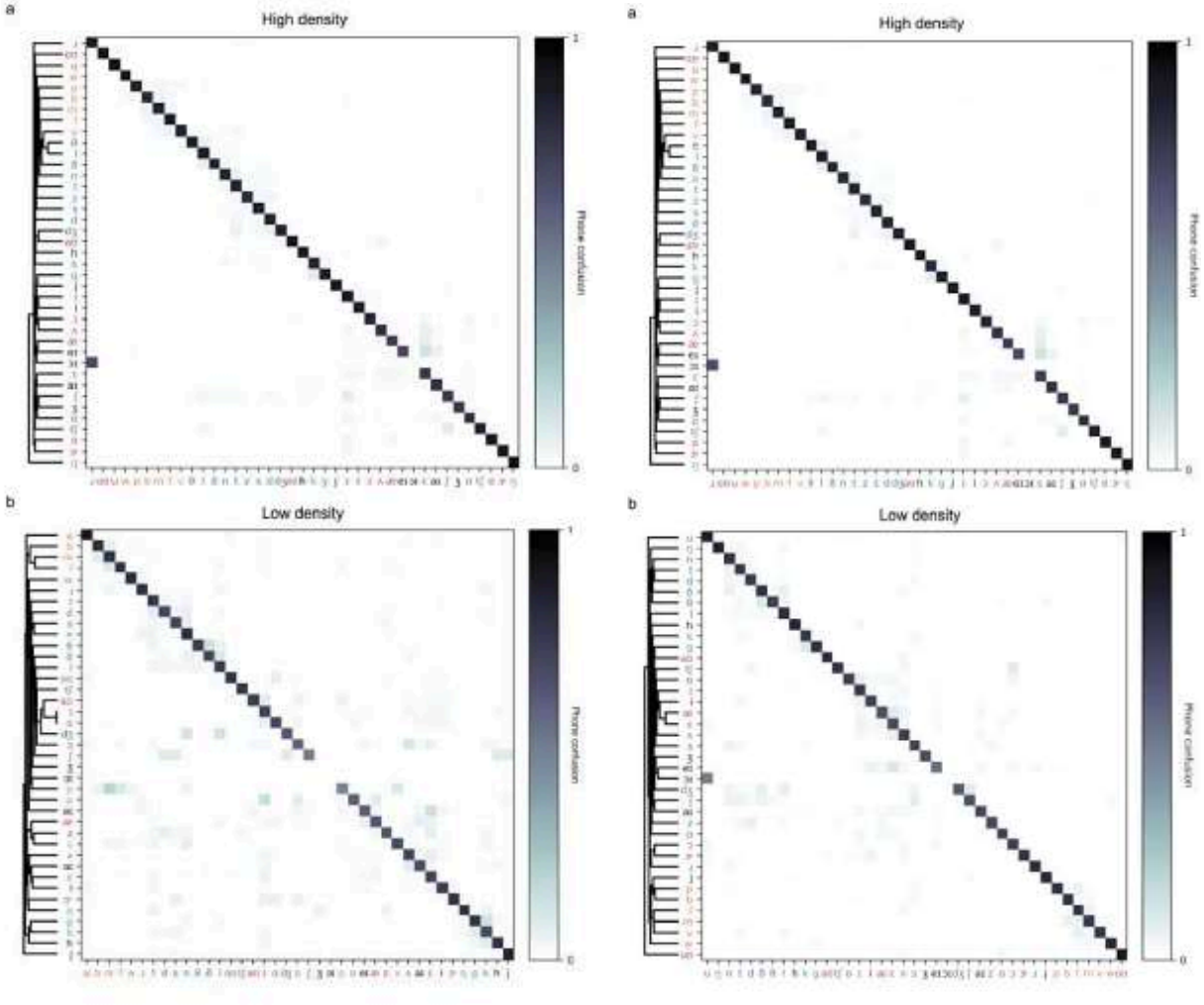


Figure S13

Depicted mean electrode-contribution curves now use the corrected anatomical assignments when determining which electrodes to include in each mean curve. General trends remain unchanged.

New figure is on the left; old figure is on the right

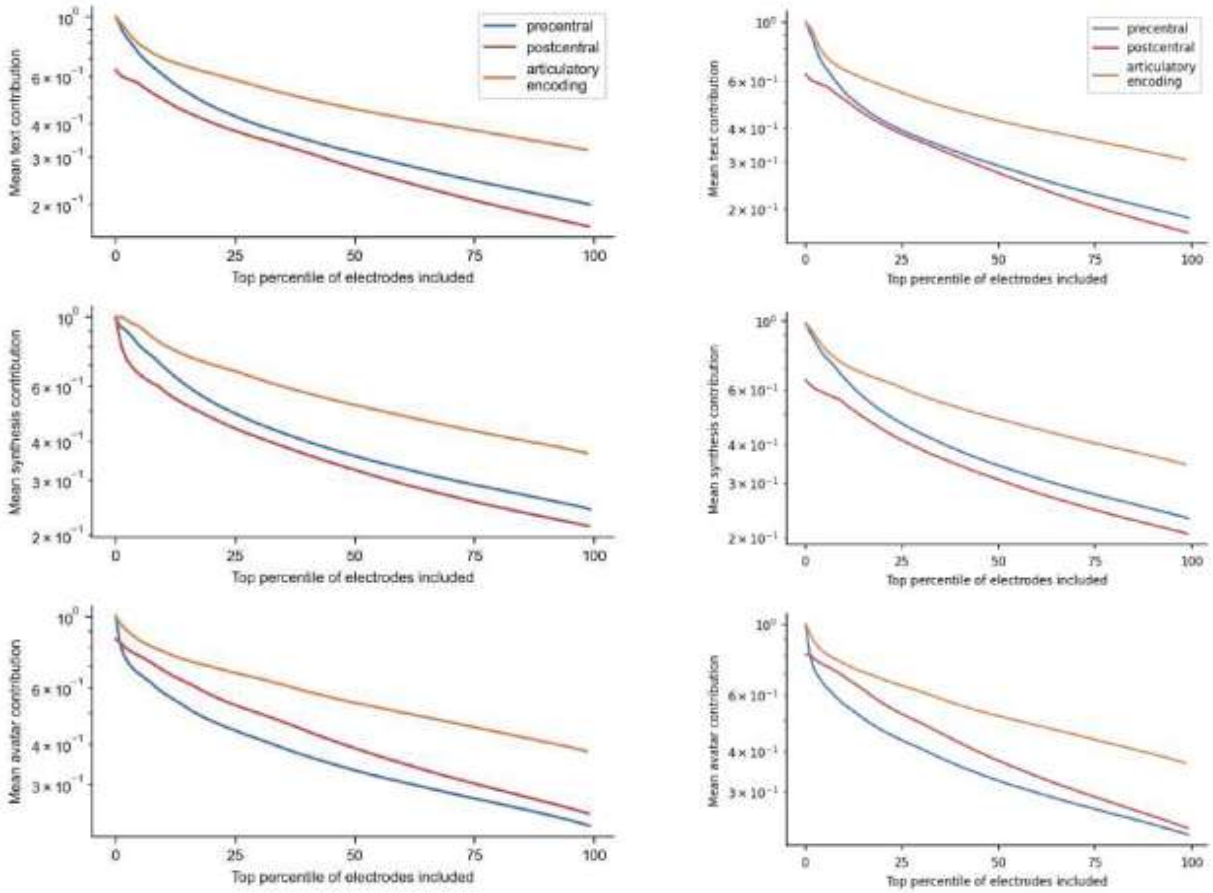


Figure S14

NATO code-word classification accuracy as a function of excluded anatomical region has been updated using corrected anatomical assignments for each electrode. Exclusion of postcentral gyrus (PoCG), precentral gyrus (PrCG), and sensorimotor cortex (SMC) electrodes now show greater performance degradations when using the corrected anatomical assignments.

The text "** P < 0.005, *** P < 0.001," was added to the caption to describe the new significance markers that appear in the revised figure.

New figure is on the top; old figure is on the bottom

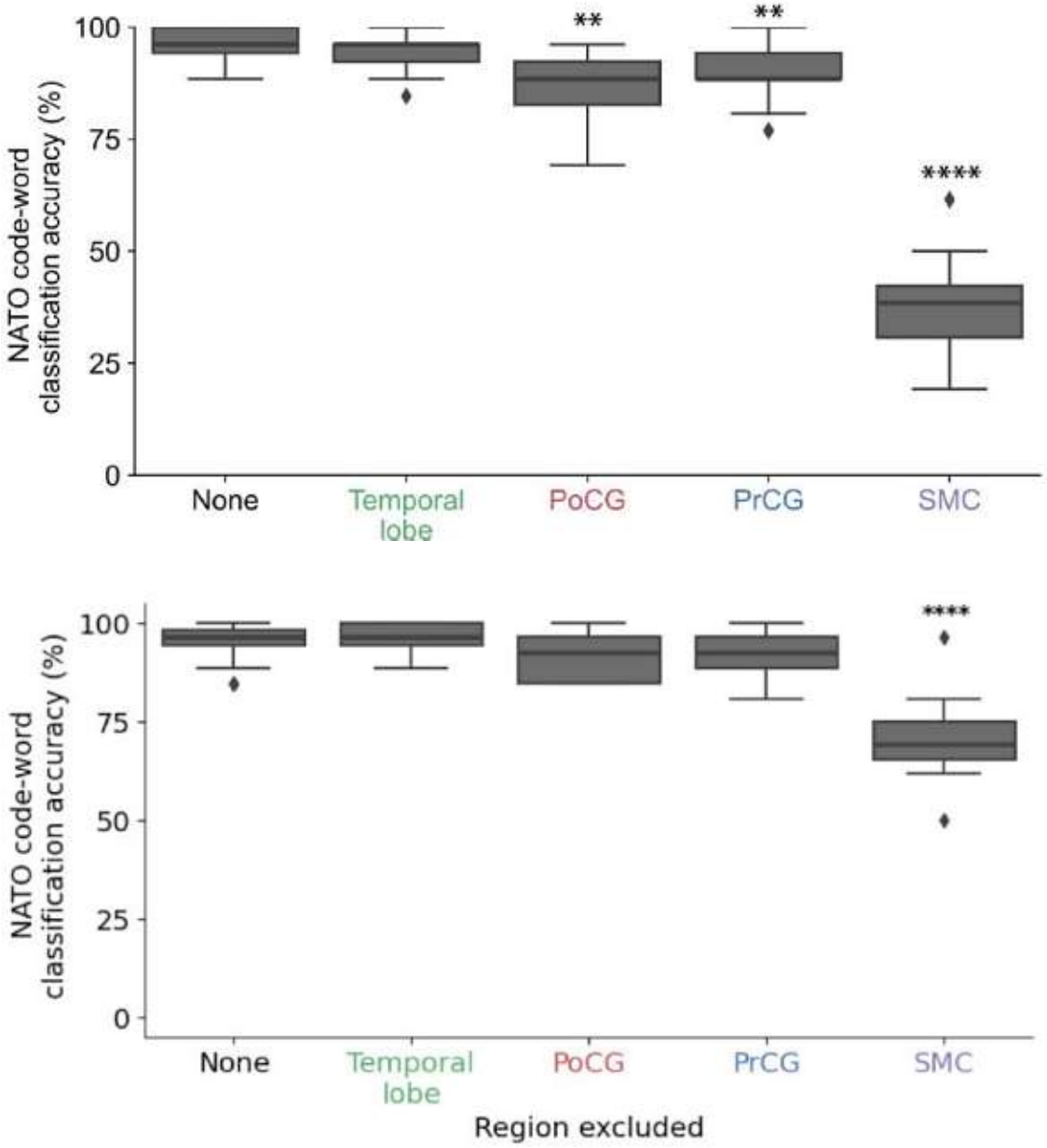


Figure S15

This set of analyses comparing electrodes from the precentral and postcentral gyri has been updated using the corrected anatomical assignments. Outcomes did not change significantly.

New figure is on the top; old figure is on the bottom

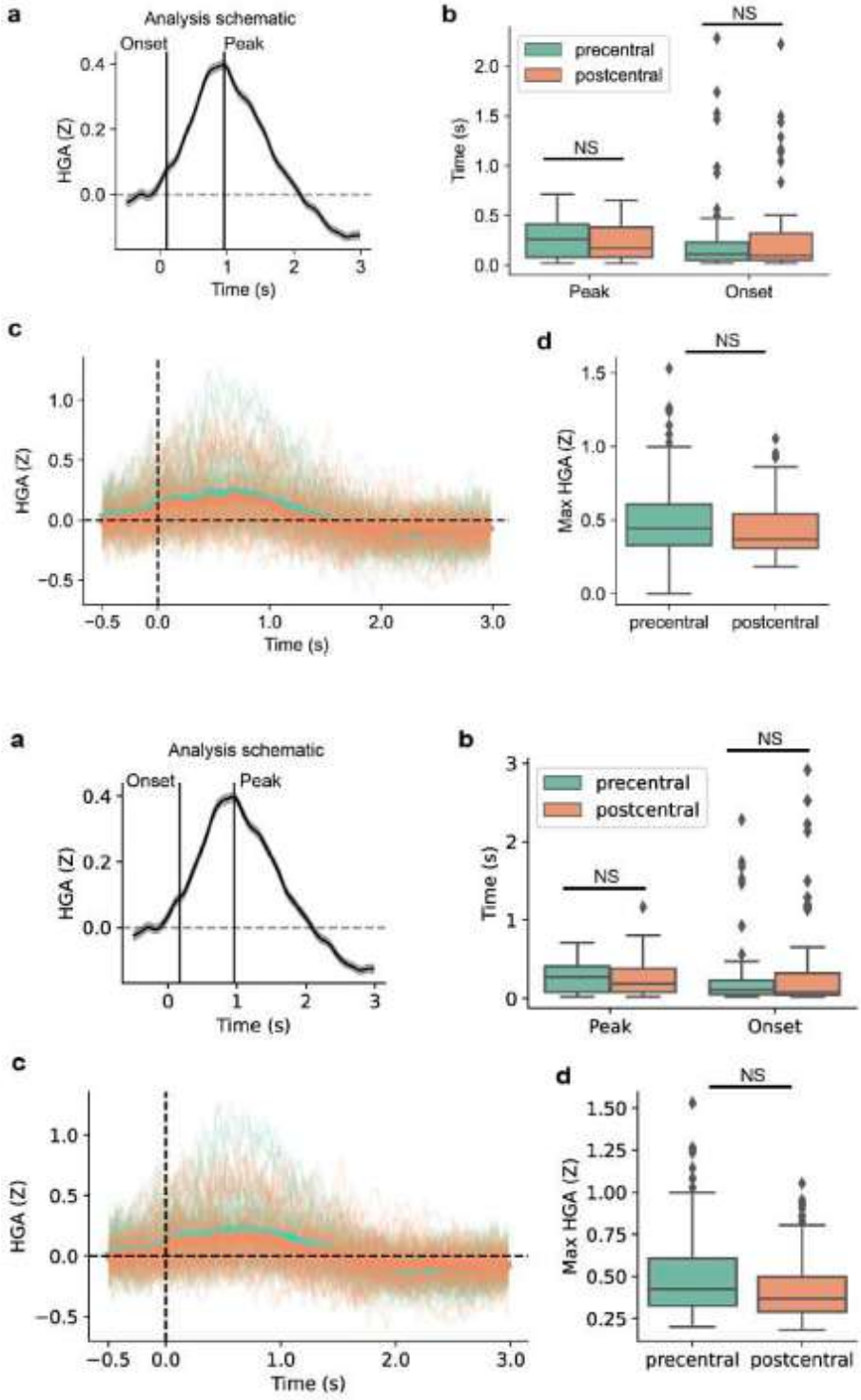


Figure S16

The relationships between evoked HGA during listening and contribution to text, synthesis, and avatar models have been re-calculated for temporal-lobe electrodes using the corrected anatomical assignments. Correlations were significantly affected and a sentence in the figure caption has been updated.

New text is on the top; old text is on the bottom

There are significant correlations between HGA during listening and electrode contribution to text-decoding (left; r = 0.55, P < 0.0001), speech-synthesis (middle; r =0.60, P < 0.0001), and avatar direct-decoding (right; r = 0.60, P < 0.0001).

There is no significant correlation between HGA during listening and electrode contribution to text-decoding (left; r = 0.017, P = 0.91), speech-synthesis (middle; r = -0.0052, P = 0.99), and avatar direct-decoding (right; r = 0.089, P = 0.51).

New figure is on the top; old figure is on the bottom

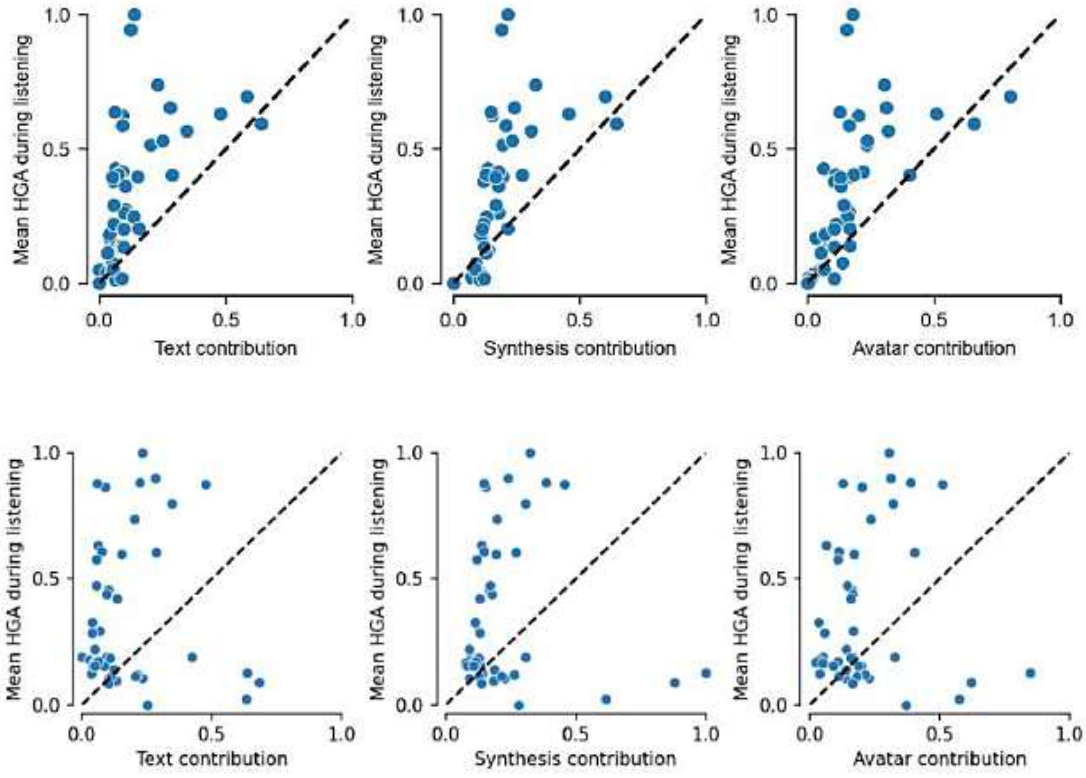


Table S7

Statistical values computed as part of region-exclusion analyses, in which decoder performance was assessed while excluding electrodes from certain anatomical regions, have been updated after repeating the analyses with the corrected anatomical assignments for the electrodes.

Shared columns with new values are on the left; old values are on the right

Task[1]	Region excluded 1	Region excluded 2	Statistic[2]	Corrected P-value[3]	Statistic[2]	Corrected P-value[3]
Text WER	SMC	temporal lobe	0.00e+00	8.94e-07	0.00e+00	8.94e-07
Text WER	SMC	chance	2.00e+00	1.07e-06	8.00e+00	1.04e-05
Text WER	SMC	none	0.00e+00	8.94e-07	0.00e+00	8.94e-07
Text WER	SMC	postcentral	0.00e+00	8.94e-07	0.00e+00	8.94e-07
Text WER	SMC	precentral	0.00e+00	8.94e-07	0.00e+00	8.94e-07
Text WER	temporal lobe	chance	0.00e+00	8.94e-07	0.00e+00	8.94e-07
Text WER	temporal lobe	none	1.31e+02	5.87e-01	8.90e+01	1.65e-01
Text WER	temporal lobe	postcentral	3.00e+00	1.49e-06	9.10e+01	1.65e-01
Text WER	temporal lobe	precentral	1.70e+01	3.70e-05	7.30e+01	5.89e-02
Text WER	chance	none	0.00e+00	8.94e-07	0.00e+00	8.94e-07
Text WER	chance	postcentral	0.00e+00	8.94e-07	0.00e+00	8.94e-07
Text WER	chance	precentral	0.00e+00	8.94e-07	0.00e+00	8.94e-07
Text WER	none	postcentral	1.00e+00	8.94e-07	2.40e+01	2.63e-03
Text WER	none	precentral	1.20e+01	1.67e-05	1.10e+01	4.29e-04
Text WER	postcentral	precentral	9.80e+01	1.70e-01	9.50e+01	3.06e-01
Synthesis MCD	SMC	temporal lobe	0.00e+00	2.86e-05	0.00e+00	2.86e-05
Synthesis MCD	SMC	chance	3.30e+01	2.23e-02	0.00e+00	2.86e-05
Synthesis MCD	SMC	none	0.00e+00	2.86e-05	0.00e+00	2.86e-05
Synthesis MCD	SMC	postcentral	0.00e+00	2.86e-05	1.00e+00	2.86e-05
Synthesis MCD	SMC	precentral	0.00e+00	2.86e-05	6.00e+00	1.07e-04
Synthesis MCD	temporal lobe	chance	0.00e+00	2.86e-05	0.00e+00	2.86e-05
Synthesis MCD	temporal lobe	none	5.00e+00	1.14e-04	1.00e+00	2.86e-05
Synthesis MCD	temporal lobe	postcentral	1.03e+02	9.56e-01	9.40e+01	7.01e-01
Synthesis MCD	temporal lobe	precentral	5.50e+01	1.27e-01	4.80e+01	6.55e-02
Synthesis MCD	chance	none	0.00e+00	2.86e-05	0.00e+00	2.86e-05
Synthesis MCD	chance	postcentral	0.00e+00	2.86e-05	0.00e+00	2.86e-05
Synthesis MCD	chance	precentral	0.00e+00	2.86e-05	0.00e+00	2.86e-05
Synthesis MCD	none	postcentral	6.00e+00	1.34e-04	0.00e+00	2.86e-05
Synthesis MCD	none	precentral	0.00e+00	2.86e-05	0.00e+00	2.86e-05
Synthesis MCD	postcentral	precentral	4.60e+01	7.99e-02	3.20e+01	1.46e-02
NATO code-word accuracy	SMC	temporal lobe	0.00e+00	3.81e-05	0.00e+00	3.81e-05
NATO code-word accuracy	SMC	none	0.00e+00	3.81e-05	0.00e+00	3.81e-05
NATO code-word accuracy	SMC	postcentral	0.00e+00	3.81e-05	0.00e+00	3.81e-05
NATO code-word accuracy	SMC	precentral	0.00e+00	3.81e-05	0.00e+00	3.81e-05
NATO code-word accuracy	temporal lobe	none	0.00e+00	1.38e-02	1.50e+01	7.15e-01
NATO code-word accuracy	temporal lobe	postcentral	0.00e+00	1.68e-03	8.00e+00	1.35e-02
NATO code-word accuracy	temporal lobe	precentral	2.50e+00	7.66e-03	8.50e+00	2.01e-02
NATO code-word accuracy	none	postcentral	0.00e+00	1.68e-03	7.00e+00	3.21e-02
NATO code-word accuracy	none	precentral	0.00e+00	2.55e-03	5.00e+00	1.25e-02
NATO code-word accuracy	postcentral	precentral	3.60e+01	1.71e-01	5.40e+01	7.28e-01
Avatar DTW mean correlation	SMC	temporal lobe	3.42e+03	9.50e-05	5.09e+03	7.37e-01
Avatar DTW mean correlation	SMC	chance	2.46e+02	1.41e-23	3.14e+02	5.11e-23
Avatar DTW mean correlation	SMC	none	3.30e+03	3.76e-05	1.54e+03	3.96e-14
Avatar DTW mean correlation	SMC	postcentral	4.32e+03	3.14e-02	4.82e+03	4.07e-01
Avatar DTW mean correlation	SMC	precentral	4.38e+03	3.14e-02	4.82e+03	4.07e-01
Avatar DTW mean correlation	temporal lobe	chance	4.80e+01	4.18e-25	0.00e+00	1.62e-25
Avatar DTW mean correlation	temporal lobe	none	4.30e+03	3.14e-02	4.07e+03	9.30e-03
Avatar DTW mean correlation	temporal lobe	postcentral	4.32e+03	3.14e-02	5.36e+03	1.00e+00
Avatar DTW mean correlation	temporal lobe	precentral	4.96e+03	2.31e-01	5.76e+03	1.00e+00
Avatar DTW mean correlation	chance	none	4.80e+01	4.18e-25	4.80e+01	3.90e-25
Avatar DTW mean correlation	chance	postcentral	2.03e+02	6.78e-24	1.81e+02	4.45e-24
Avatar DTW mean correlation	chance	precentral	1.40e+02	2.18e-24	5.10e+01	3.90e-25
Avatar DTW mean correlation	none	postcentral	3.90e+03	3.08e-03	2.30e+03	8.65e-10
Avatar DTW mean correlation	none	precentral	3.65e+03	5.67e-04	3.39e+03	6.48e-05
Avatar DTW mean correlation	postcentral	precentral	5.22e+03	2.74e-01	5.75e+03	1.00e+00

Table S8

Statistical values computed as part of electrode-density analyses, in which decoder performance was assessed while simulating different electrode densities, have been updated after repeating the analyses with the corrected anatomical arrangement of the electrodes.

Shared columns with new values are on the left; old values are on the right

Task[1]	Density 1	Density 2	Statistic[2]	P-value[3]	Statistic[2]	P-value[3]
Text WER	chance	high density	0.00e+00	1.79e-07	0.00e+00	1.79e-07
Text WER	chance	low density	0.00e+00	1.79e-07	0.00e+00	1.79e-07
Text WER	high density	low density	5.00e+00	5.96e-07	2.40e+01	3.18e-04
Synthesis MCD	chance	high density	0.00e+00	5.72e-06	0.00e+00	5.72e-06
Synthesis MCD	chance	low density	0.00e+00	5.72e-06	0.00e+00	5.72e-06
Synthesis MCD	high density	low density	1.20e+01	1.34e-04	9.00e+00	6.29e-05
Avatar DTW mean correlation	chance	high density	4.80e+01	8.36e-26	4.80e+01	5.57e-26
Avatar DTW mean correlation	chance	low density	6.00e+01	8.36e-26	1.90e+01	4.72e-26
Avatar DTW mean correlation	high density	low density	5.00e+03	1.34e-01	4.34e+03	6.59e-03